

Article

Effects of different nozzle materials on atomization results via CFD simulation

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ABSTRACT

Spray drying, as a crucial operation in industrial production, converts solution to fine particle. The spray moiety directly affects the final particle morphology, size and distribution. Compared with the experimental method, computational fluid dynamics (CFD) modeling is a powerful and convenient tool for simulating the spray process. Based on the verified CFD model, different materials of atomizer were simulated to investigate the effect on droplet size and distribution in this work. The modeling result proved that the droplet size and distribution were influenced by the resistance coefficient of materials, wherein the Reynolds number could change the effect of roughness along with the change of mass flow rate on spray process. The results in this work have implication for controlling droplet size through developing new spray nozzle with different materials or surface coating.

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1. Introduction

Spray drying plays a very important role in many manufacturing processes and industrial areas, such as catalyst producing, drug engineering and dairy industry [1–3]. The spray drying converts the slurry or solution into droplets by atomizers and evaporates solvent to obtain particles through hot gaseous drying medium [4]. During the spray drying process, the spray moiety significantly affects the particle morphology, size and distribution, which is most important index in affecting product properties greatly. For example, the diameter of drug products decides the dissolution rate in water and absorption rate in human body [2]. Besides, the dairy nutrient content is related to dairy product size and distribution, which is one of the most important indexes [3]. However, the spray is a complicated process which was affected by various factors, including solution physicochemical characteristics (concentration, surface tension, viscosity, and density), process parameters (mass flow rate, air pressure, temperature and spray angle) and nozzle structure (orifice diameter and contact types between gas and liquid) [5]. It requires a large amount of time and cost to explore the optimized condition of spray through traditional experimental methods. Hence, numerical simulation methods were developed and introduced to study the engineering issues due to advantages of preciseness and convenience.

The computational fluid dynamics (CFD) is a vital branch of numerical simulation to solve the governing equation of fluid dynamics which was utilized to guide and predict the experimental research efficiently by computer and numerical computation, when the modeling results were well corresponded with experiment results [6–8]. Therefore, various models have been built based on CFD simulation method which was applied for spray process by a combinative way of modeling and experiment. Rizkalla *et al.*, for example, proved that the increase of liquid density led to an increase in droplet size [9]. In addition, the viscosity and surface tension usually influenced spray process jointly. Ejim *et al.* reported that the increase of the hydroxypropyl cellulose concentrations resulted in an increase of droplet diameter which could be a relation of both viscosity and surface tension [10]. For process parameters, the droplet size increased with mass flow rate, while the increase of air pressure or spray angle induced a reduction on droplet diameter [11,12]. Especially, the change of air pressure influenced the relative velocity between gas and liquid, as the most direct and effective factor for the droplet size and distribution [13]. In recent, the nozzle structure factor attracted much attention. For instance, Zhang *et al.* investigated how the diameter of nozzle orifices affected droplet size distribution [14]. As a result, the optimum nozzle orifice length/diameter ratio was supposed to be 2 for different nozzle types. Elkotb *et al.* found that the external-mixing air-blast atomizers had an energy loss in the orifice, while the internal-mixing air-blast atomizers needed a complex mixing process for gas and liquid [15].

Actually, the different atomizer materials possess different properties, such as resistance coefficient, surface physicochemical

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characteristics and roughness degree, which have great influence on the spray result. For example, resistance coefficient resulted in different relative velocities between gas and liquid, which affect the droplet size and distribution obviously. Although different materials were applied to produce nozzles, the investigation of the effect on droplet size has not been reported. In this work, herein, a verified CFD simulation model was built to investigate droplet size and distribution using air atomizing nozzle with three materials: aluminum, brass, and stainless-steel. The droplet diameters were significantly different at low mass flow rates for these three materials, while the diameters tended to be uniform with the increase of mass flow rates. The result in this work revealed that the droplet size and distribution were influenced by the resistance coefficient of materials, wherein the Reynolds number could affect roughness along with the change of mass flow rate on spray process. Furthermore, the results in this work can be applied to control the particle size and distribution in product manufacturing, for example, by using different material nozzle and/or coating nozzle with various materials.

2. Theoretical Background

The CFD model needs to describe the flow of liquid and gas, the formation of droplet and the exchange of momentum and so forth. All the processes are physical change without chemical reactions. Hence, some basic physical equations are set out below.

The equation of mass conservation is also called continuous equation [16]. The common form for mass conservation equation in Eq. (1) can apply to both compressible and incompressible flow. The source item is the mass that transfer from discrete phase to continuous phase included the other definition of mass source item.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = S_m \quad (1)$$

Actually, the momentum conservation equation is the Newton's second motion law. In the inertial reference system, the momentum conservation can be demonstrated by the following expression:

$$\frac{\partial}{\partial t} (\rho \vec{v}) + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla P + \nabla \cdot (\tau) + \rho \vec{g} + \vec{F} \quad (2)$$

The parameter P is static pressure, $\rho \vec{g}$ means gravity. $\vec{F}$ contains two parts: the external force and the source item based on other models. τ is pressure tensor which can be characterized as follow:

$$\tau = \mu \left[(\nabla \vec{v} + \nabla \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (3)$$

In the equation, the parameter μ is molecular viscosity and I is unit tensor. The second source item means the influence from volume expansion.

Rigorous energy conservation includes the energy transfer in fluid flow (Bernoulli equation) and the heat transfer which is written as [17]:

$$\frac{\partial}{\partial t} (\rho E) + \nabla \cdot [\vec{v} (\rho E + p)] = \nabla \cdot [K_{eff} \nabla T - \sum_j h_j \vec{J}_j + (\vec{\tau}_{eff} \cdot \vec{v})] + S_h \quad (4)$$

wherein, K_{eff} is the effective conductivity and $\vec{J}_j$ is the diffusion flux of species j (gas-phase). The source item (S_h) on the right of the equation standards for the heat conduction, component diffusion and viscosity dissipation. The S_h contains heat source from chemical reaction and others. In Eq. (4), the parameter E can be calculated through Eq. (5):

$$E = h - \frac{p}{\rho} + \frac{v^2}{2} \quad (5)$$

wherein, h represents for enthalpy, for ideal gas:

$$h = \sum_j Y_j h_j \quad (6)$$

for incompressible liquid:

$$h = \sum_j Y_j h_j + \frac{p}{\rho} \quad (7)$$

In the Eqs. (6) and (7), Y_j is the mass fraction of species j , and the enthalpy of species j can be calculated by:

$$h_j = \int_{T_{ref}}^T c_{pj} dT \quad (8)$$

The parameter T_{ref} is the reference temperature in calculating the enthalpy of different components. Different solvers and models have different reference temperature in the modeling software FLUENT [16].

The heat and mass transfer coefficient between continuous phase and discrete phase are calculated by well-known Ranz–Marshall equation [18]. The coefficient of mass transfer is described as follows:

$$Nu = \frac{h d_p}{\kappa_\infty} = \left(2 + 0.6 Re^{0.5} + Sc^{0.33} \right) \quad (9)$$

The coefficient of heat transfer is calculated as follows:

$$Sh = \frac{h_m d_p}{D_i} = \left(2 + 0.6 Re^{0.5} + Pr^{0.33} \right) \quad (10)$$

Since the process of drying was not involved in this work, the evaporation model, moisture model, and shrinkage model are not studied in the simulation.

3. Geometry Description

Geometry figure of the air atomizing nozzle and simplified schematic of the calculation domain with an intensive region of the mesh density were shown in Fig. 1. To simplify the modeling process and avoid potential instability of CFD simulation, the 3D geometry model only contained the orifice part of nozzle. The nozzle used in this work comes from Spraying System Co (SUV67B from Spraying System Co), which is comprised of three parts: the central orifice, the assistant orifices and the sector controlling orifices. At the middle of the nozzle, the inner diameter of central orifice is 4.6 mm. Because of the low mass flow of assistant orifice, the effect of assistant orifices is negligible and it was not considered in this 3D geometry. The diameter of sector controlling orifices is 1.2 mm and the spray pattern would be ellipse with opened sector controlling orifices and there would be different ovalities for different pressures. The calculation domain was a cylinder with 1.5 m in height and 0.5 m in diameters. The solution was sprayed in the area of cylinder.

The computational mesh was created by ICEM. Because the inlet and outlet are the core section of simulation, the high-density mesh was adopted in these regions. The calculation domain contained two parts: the orifice of air atomizing nozzle and spray flow field. After considering the spray pattern and convenience of calculation, the calculation domain was simplified to a column with diameters of 0.8 m and 1.5 m. Referring to the work by Yang *et al.* [19] about analyzing the independence of mesh number (Fig. S1), the final mesh contains 78,178 nodes and 439,068 tetrahedral cells in total.

4. Simulation Strategy

The 3D double-precision solver for the simulation was utilized in ANSYS FLUENT v.19 [16]. The software used the finite volume discretization as the default method. The discretization scheme of

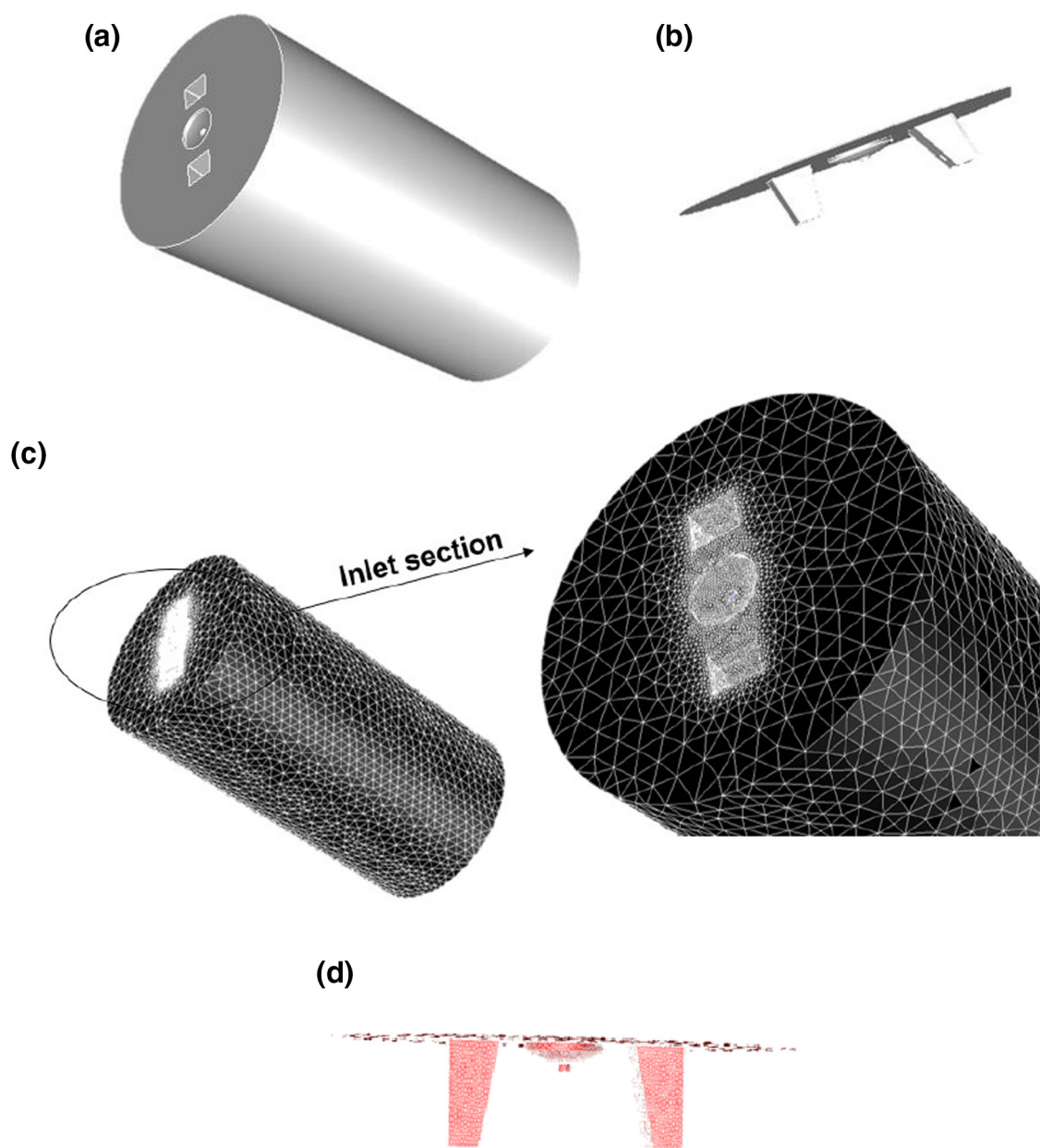


Fig. 1. (a) Schematic diagram of the calculation domain in FLUENT and (b) the inlet section of model, (c) calculation mesh used in the simulation and a magnified view of the inlet section with high-density mesh. (d) The calculation mesh of orifice used in the simulation.

second-order upwind was applied to the convective terms, and central differencing scheme was used for the diffusion terms. As to the turbulent model, Trujillo [20] found that the simulation results with RNG $k-\epsilon$ model matched greatly with experimental results, therefore, it was chosen for the modeling process. SIMPLE consistent scheme for the pressure–velocity coupling and PRESTO! for the pressure interpolation were used to improve the efficiency and accuracy.

The procedure of the transient simulation was step by step. Jin and Chen [21] compared the step size of 0.01 s and 0.001 s in a 3D spray modeling, and the result showed that there is no significant difference. A comparison of various step sizes (0.01 s, 0.001 s, 0.005 s) showed that the model result was almost the same. Hence, the step size of 0.01 s was chosen in order to save simulation time.

The discrete phase model, proposed by Crowe [22] and Smoot [23], was based on the Eulerian–Lagrangian method. Therefore, the droplet collision model, breakup model and the change of drag coefficient were studied for modeling spray process. The droplet collision model

was more suitable for the collision with low weber number, and the result of collision would be combined or rebounded. For high weber number (more than 100) in this modeling, the collision model was ignored for accurate result. As to the breakup model, the Taylor's analogy breakup model (TAB) [24] is suitable for low speed and low weber number process. Therefore, the WAVE model, proposed by Hwang and Reitz [25] was a better choice. The precision of drag coefficient in FLUENT was crucial for accurate simulation and applicable for all the situation. Hence, the WAVE model and the dynamic-drag model were available in the modeling process.

5. Inlet, Outlet and Boundary Conditions

The calculation domain contains three pressure inlets and one pressure outlet. The wall material was set as the aluminum at first. The experiment and simulation were carried out at 25 °C. The mixture was dissolved in the ethyl acetate with different mass proportion (10%

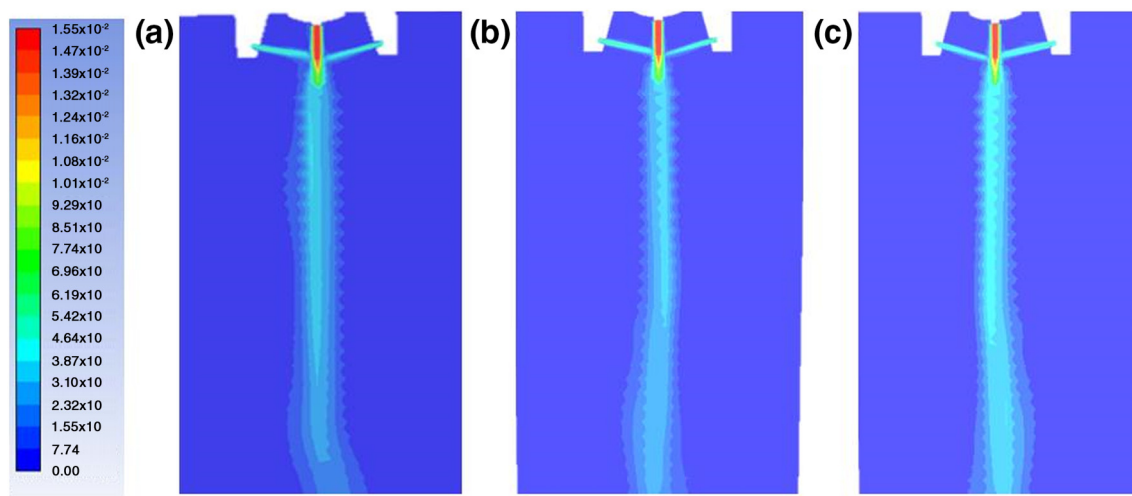


Fig. 2. Contours of velocity in axial cross-section at different mass flow rates ($\text{m}\cdot\text{s}^{-1}$): (a) $20\text{ L}\cdot\text{h}^{-1}$, (b) $30\text{ L}\cdot\text{h}^{-1}$, (c) $50\text{ L}\cdot\text{h}^{-1}$.

and 20%) to simulate the influence of viscosity and surface tension. The viscosity was tested by Ubbelohde viscometer and the surface tension was tested by tensiometer (KRUS100C). The calculated quantities of turbulence kinetic energy (k) and the turbulent dissipation rate (ϵ) for the air inlet were $144\text{ m}^2\cdot\text{s}^{-2}$ and $797,850\text{ m}^2\cdot\text{s}^{-3}$, respectively. The influence of mass flow rate and spray angle were also considered in the DPM.

6. Results and Discussion

6.1. Model validation

The continuous modeling result was shown in Fig. 2 with different mass flow rates. The velocity contours were almost the same, meaning that the continuous phase modeling was not affected by the mass flow. Therefore, the discrete phase was simulated based on these continuous models.

6.1.1. Different mass flow rates of solutions

In order to validate the spray model, CFD simulations were conducted under different mass flow rates ($20\text{ L}\cdot\text{h}^{-1}$, $30\text{ L}\cdot\text{h}^{-1}$, $50\text{ L}\cdot\text{h}^{-1}$). A comparison of the modeling and experimental droplet distribution was shown in Fig. 3. The droplet distribution concentrated in $15\text{--}35\text{ }\mu\text{m}$ for

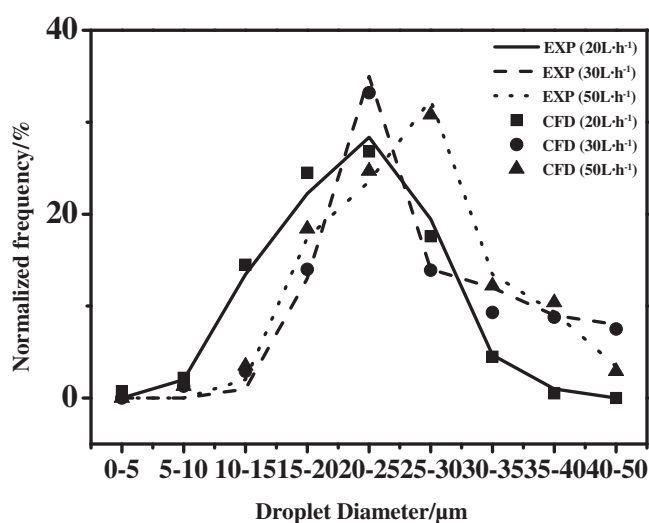


Fig. 3. Comparison of simulation results (points) and experimental results (lines) of particle size distribution with atomizing air pressure of 0.2 MPa at mass flow rates of $20\text{ L}\cdot\text{h}^{-1}$, $30\text{ L}\cdot\text{h}^{-1}$ and $50\text{ L}\cdot\text{h}^{-1}$.

both the modeling and experiment profiles. The mean diameter enlarged about $10\text{ }\mu\text{m}$ with the increase of mass flow rates from $20\text{ L}\cdot\text{h}^{-1}$ ($20.4\text{ }\mu\text{m}$, square point) to $50\text{ L}\cdot\text{h}^{-1}$ ($29.8\text{ }\mu\text{m}$, triangular point). Because the solution had not been atomized sufficiently with the increase of feed solution, the large droplet produced and the distribution of droplets with high mass flow rates was not uniform compared to droplets at low feed rates. In addition, the droplet size of simulated cases was smaller than the experimental results for about $2\text{--}5\text{ }\mu\text{m}$, because the simulation conditions were ideal compared with the experimental method. For example, the nozzle became rough in practical experiment which was not considered in the CFD simulation.

6.1.2. Different gas pressures

Gas pressure is a vital factor in the spray process which affects the relative velocity between gas and liquid effectively and directly. In this section, different pressures of atomizing gas (0.2 MPa , 0.3 MPa , 0.4 MPa , 0.5 MPa) were used for the simulation with constant mass flow rate of $30\text{ L}\cdot\text{h}^{-1}$.

As shown in Fig. 4, the droplet size distribution followed the Poisson distribution at 0.2 MPa and 0.3 MPa , while the mean diameter decreased about $10\text{ }\mu\text{m}$ with more concentrated distribution when the air pressure was enhanced from 0.2 MPa to 0.3 MPa . As expected, the results were attributed to the high atomization pressure induced high relative velocity between gas and liquid. According to the atomization theory [25], the droplet split into fine droplet with the aerodynamic effect in the second atomization process. However, only part of simulated droplet distribution over $6\text{--}10\text{ }\mu\text{m}$ was obtained when the gas pressure increased to $0.4\text{--}0.5\text{ MPa}$. The high gas pressure induced high energy, leading to the contravention of Poisson distribution. Therefore, the air pressure was set up over the range of $0.2\text{--}0.3\text{ MPa}$ in the following simulation.

6.1.3. Different concentrations of feed solution

The droplet size distribution was investigated by increasing the solution concentration from 10% to 20% (w/w). The change of viscosity and surface tension was found accompanied with solution concentrations. The influence of viscosity and surface tension was usually considered together in the modeling process [10]. The viscosity and surface tension values of solutions were measured as shown in Fig. S4. The viscosity of 20% solution (w/w) was about $55\text{ mPa}\cdot\text{s}$ which was significantly higher than that for 10% solution ($20\text{ mPa}\cdot\text{s}$) at $25\text{ }^\circ\text{C}$. Along with the increase of temperature from $25\text{ }^\circ\text{C}$ to $65\text{ }^\circ\text{C}$, the difference of viscosity diminished from $35\text{ mPa}\cdot\text{s}$ to $10\text{ mPa}\cdot\text{s}$. The disparity of the surface tension was only $0.7\text{ mN}\cdot\text{m}^{-1}$ between 10% and 20% solution at $25\text{ }^\circ\text{C}$.

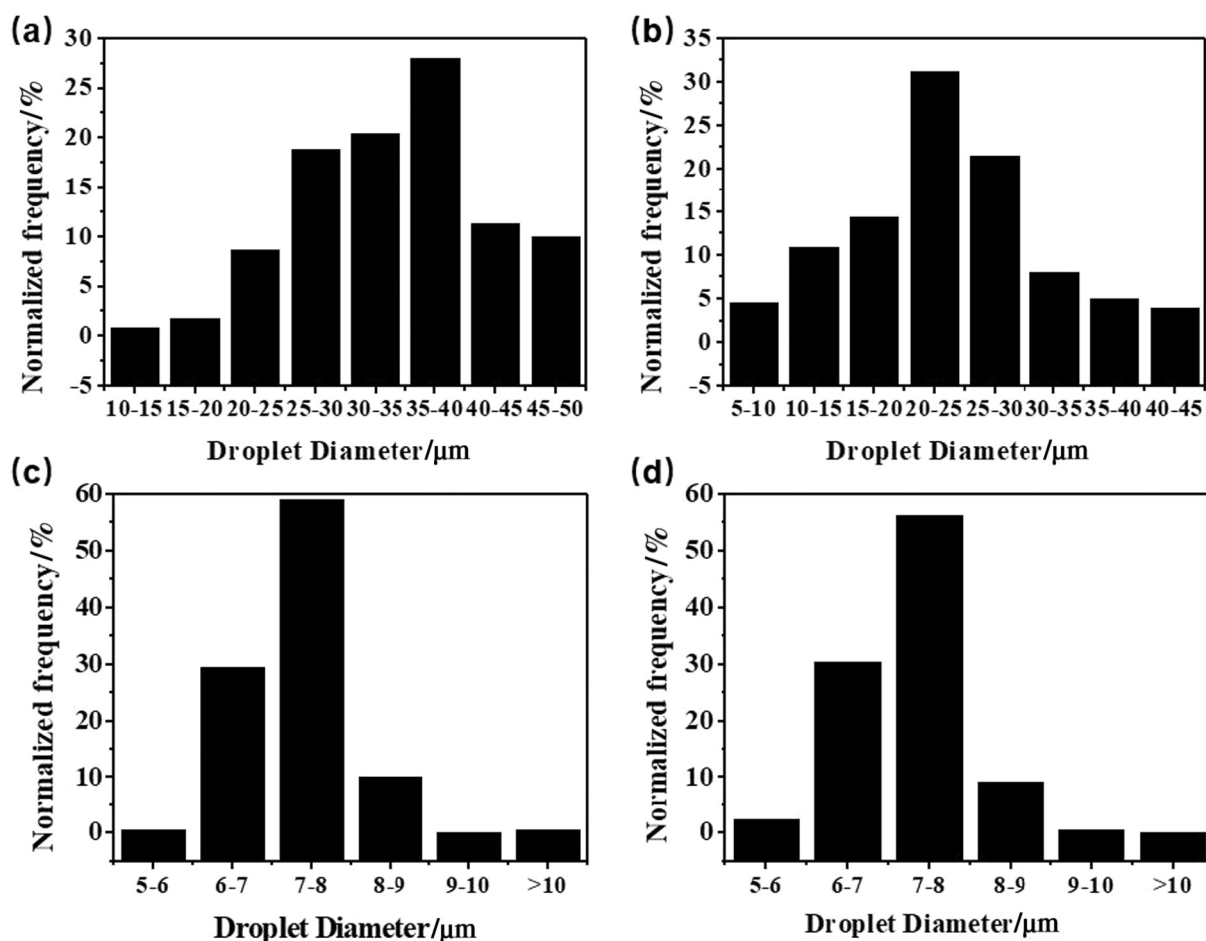


Fig. 4. Simulation results of particle size distribution with mass flow rates at $30 \text{ L} \cdot \text{h}^{-1}$ under air pressures of (a) 0.2 MPa, (b) 0.3 MPa, (c) 0.4 MPa, and (d) 0.5 MPa.

The simulation results of different concentrations showed great agreement with the experimental data on droplet size distribution in Fig. 5. The distribution curves were similar, though peak values of droplet size decreased about $5 \mu\text{m}$ when the solution concentration increased from 10% (square point) to 20% (rounded point) (w/w). According to the established rules concerning the effect of viscosity and surface tension on the droplet size distribution [22], the reason of above diameter reduction was due to the increased ability of liquid in resisting the aerodynamic force of atomization. The resistance of

aerodynamic force enhanced along with the increase of viscosity and surface tension, resulting in the enlargement of droplet size.

The comparisons of experimental results with modeling simulation using different mass flow rates, different air pressures and different solution concentrations showed that the model developed in this work could be applied to perform reliable simulations of spray process.

6.2. Influence of different atomizer materials

When the droplet size distribution was monomodal, the mean diameter of droplets was used to reflect the droplet distribution range in the following research. At first, the spray process with different mass flow rates was simulated using three kinds of materials (brass, stainless steel, and aluminum) under atomizing air pressure of 0.2 MPa. In the low mass flow rate modeling ($20 \text{ L} \cdot \text{h}^{-1}$), as shown in Fig. 6a, the simulated results of the mean diameter were obviously different ($19.8 \mu\text{m}$, $22.9 \mu\text{m}$, and $24.3 \mu\text{m}$) when nozzle materials were brass, stainless steel, and aluminum respectively. However, three mean diameters were tended to be identical along with the increase of mass flow rates ($50 \text{ L} \cdot \text{h}^{-1}$). Therefore, the properties of nozzle materials would affect the droplet size at low mass flow rates under same simulating condition.

Roughness height is one of main indexes for different materials. Inspired of roughness effect on the air flow inside the drying chamber [26], three different roughness heights (0.0015 mm, 0.02 mm, 0.8 mm) were set in the boundary condition with roughness constant of 0.5. The simulation results of mean diameter with different mass flow rates and the size distribution at $20 \text{ L} \cdot \text{h}^{-1}$ were obtained (Fig. 6b–c). As shown in Fig. 6b, the droplet size distribution showed a

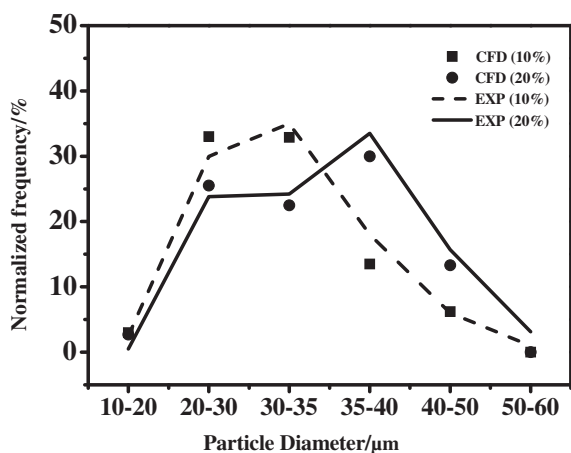


Fig. 5. Comparison of simulation result (points) and mean experimental results (lines) of particle size distribution of two different concentrations: 10% and 20% (w/w) (atomizing air pressure = 0.2 MPa, mass flow rate $M = 30 \text{ L} \cdot \text{h}^{-1}$).

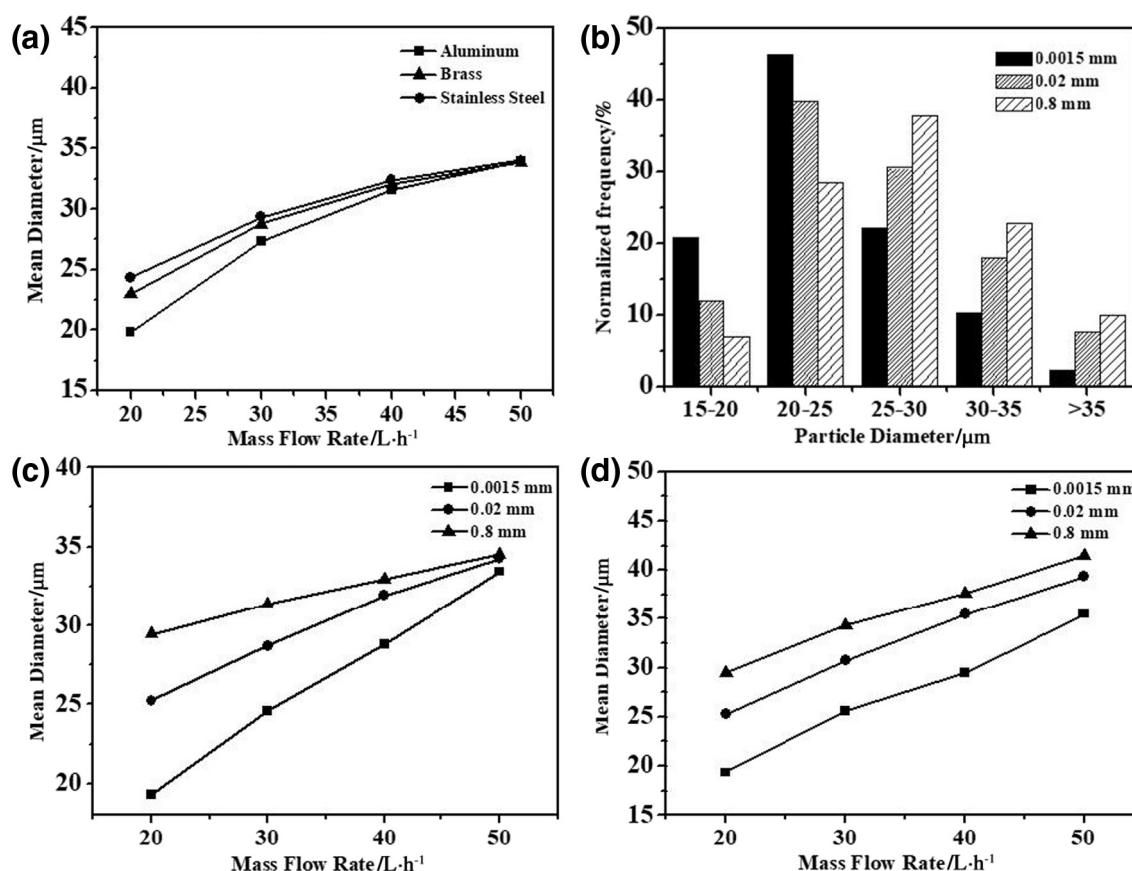


Fig. 6. Simulation results of (a) the mean diameter with different materials: aluminum, brass and stainless steel; (b) the droplet size distribution at $20 \text{ L}\cdot\text{h}^{-1}$ and (c) the mean diameter with different roughness heights: 0.0015 mm, 0.02 mm, 0.8 mm; (d) the mean diameter with constant Reynolds number (mass flow rates: $20\text{--}50 \text{ L}\cdot\text{h}^{-1}$, air pressure = 0.2 MPa).

significant difference for the three roughness heights. As to different mass flow rates in Fig. 6c, when the roughness height was changed from 0.0015 mm to 0.8 mm, the mean diameter increased from $18 \mu\text{m}$ to $30 \mu\text{m}$. With increasing mass flow rates, significant enlargements were observed on the mean diameter for the three cases. However, mean diameters for three cases reached about $35 \mu\text{m}$ when the mass flow rate reached $50 \text{ L}\cdot\text{h}^{-1}$.

As shown in Fig. 6c, both the roughness and the mass flow rate led to the change of droplet diameter. The changes of mass flow rates were essentially the change of the Reynolds number. Therefore, the resistance coefficient, which was mainly affected by roughness and Reynolds number, was supposed to be an important factor for controlling the droplet diameter. The roughness heights exhibited direct influence on the value of the resistance coefficient, while the Reynolds affected resistance coefficient indirectly through collision and curvature in the flow. Because the boundary layer of laminar flow was quite thin in the turbulent flow, the energy loss from inner collision of flow and curvature of streamline was dominated, compared with the energy loss from the divergency of laminar boundary layer. The increase of roughness heights would lead to high frequency of collision and curvature, further resulting in the increase of resistance coefficient. Furthermore, the increase of mass flow rates induced growth of Reynolds number, which made the velocity distribution uniform in the pipe with less energy loss. Therefore, resistance coefficient tended to reduce along with the increase of Reynolds number, while resistance coefficient would increase with increasing roughness. In practical, the control of droplet size could be realized through the adjustment of nozzle roughness at low mass flow rates, for example using different materials or surface coatings.

Moreover, three cases of different roughness heights were simulated with increased mass flow rate while keeping Reynolds number constant. As shown in Fig. 6d, the mean diameter of droplets increased along with the increase of mass flow rates (from $20 \text{ L}\cdot\text{h}^{-1}$ to $50 \text{ L}\cdot\text{h}^{-1}$), while the difference of mean diameter among cases with different roughness heights showed little changes with increasing mass flow rates. Thus, the obvious difference of mean diameter was caused by the different resistance coefficients which was attributed to the change of roughness heights. The smooth material with smaller resistance coefficient tended to get smaller droplet diameter and more concentrated droplet size distribution.

7. Conclusions

In this work, the effect of resistance coefficients on particle size and distribution in the spray process was studied. A nozzle model was built and the simulation results with different mass flow rates, air pressures and solution concentrations were compared with experimental data to verify the computational fluid dynamic model. The resistance coefficient of materials was determined by both roughness degree and the Reynolds number. Based on the verified model, therefore, the spray process was simulated using different atomizer materials: aluminum, brass and stainless steel. The result showed that the increase of roughness degree for different nozzle materials aggrandized the resistance coefficient directly, resulting in the obviously increase of droplet mean diameter at low mass flow rate ($20 \text{ L}\cdot\text{h}^{-1}$). Along with the increase of mass flow rate, all the droplet sizes enlarged with different rates and became identical when the mass flow rate reached $50 \text{ L}\cdot\text{h}^{-1}$. However, all the droplet sizes enlarged with the similar rates with increasing mass flow rates when the Reynolds number was constant. The results

would be great of utilization for the size control in the spray process using different nozzle materials.

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Nomenclature

c_{pj}	the specific heat capacity of species j , $\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$
E	kinetic energy for unit time and mass, J
$\vec{F}$	the external item and source item
h_j	the enthalpy of species j , $\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$
I	unit tensor
$\vec{J}_j$	the diffusion flux of species j , $\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$
K_{eff}	the effective conductivity, $\text{S} \cdot \text{m}^{-1}$
Nu	Nusselt number, dimensionless
P	static pressure, Pa
p	pressure, Pa
Re	Reynolds number
Sc	Schmidt number
Sh	Sherwood number
S_h	source item for energy transfer equation
S_m	mass source item
T	temperature, $^{\circ}\text{C}$
T_{ref}	the reference temperature, $^{\circ}\text{C}$
Y_j	the mass fraction of species j , %
∇	Hamiltonian operator
κ	thermal conductivity, $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$
ν	diffusion volume
ρ	fluid density, $\text{kg} \cdot \text{m}^{-3}$
τ	pressure tensor

Supplementary Material

Supporting Information Available: Profiles of the mesh independence modeling by flow simulation and measurement of viscosity and surface tension. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cjche.2019.09.008>.

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